Лабораторная работа №8

Настройка сетевых сервисов DHCP и DNS

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Цель работы

Основная цель

Приобретение практических навыков настройки DHCP и DNS в локальной сети Cisco Packet Tracer

Настройка DNS-сервера

Топология сети

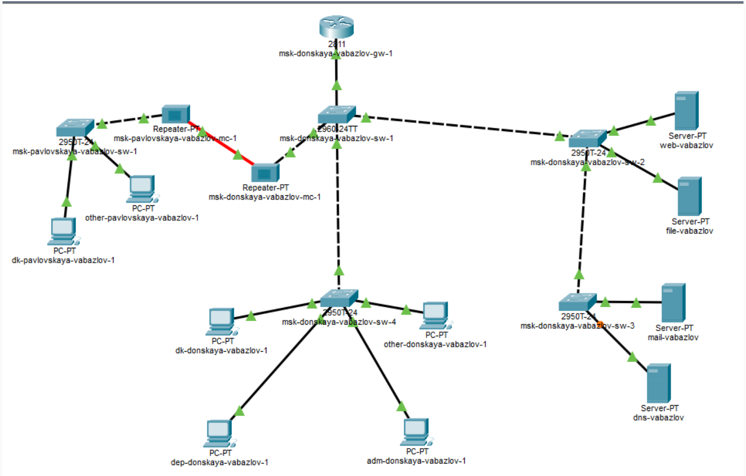


Рис. 1: Схема сети

IP-настройка сервера

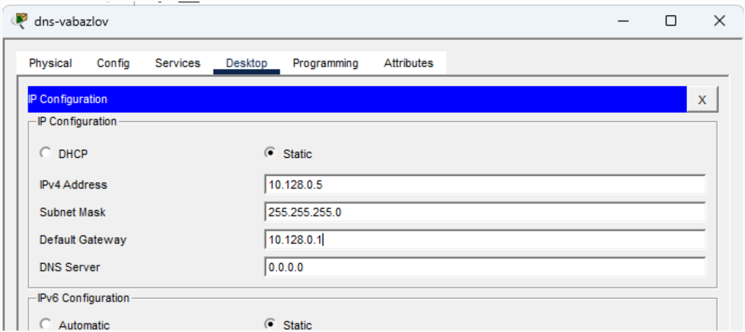


Рис. 2: IP DNS

DNS-записи

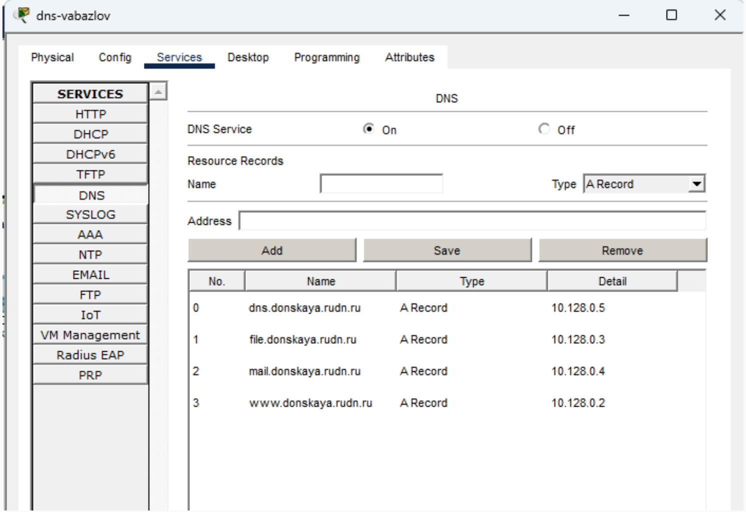
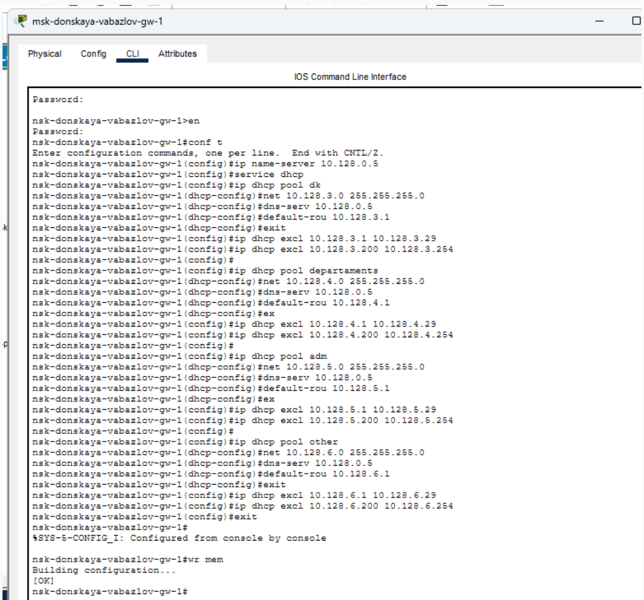


Рис. 3: DNS Records

Настройка DHCP

Конфигурация маршрутизатора



Пулы адресов

- dk — 10.128.3.0/24
- departments — 10.128.4.0/24
- adm — 10.128.5.0/24
- other — 10.128.6.0/24

Проверка DHCP

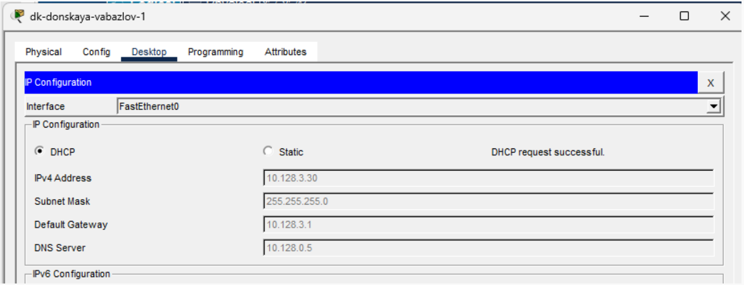


Рис. 5: DHCP pool

Таблица привязок

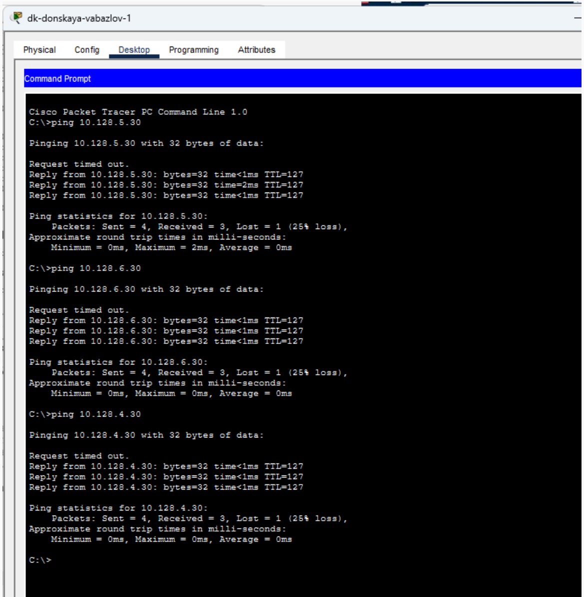
```
nsk-donskaya-vabazlov-gw-1#
nsk-donskaya-vabazlov-gw-1#sh ip dhcp pool

Pool dk :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.3.1           10.128.3.1       - 10.128.3.254      2    / 8     / 254

Pool departaments :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.4.1           10.128.4.1       - 10.128.4.254      1    / 8     / 254

Pool adm :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.5.1           10.128.5.1       - 10.128.5.254      1    / 8     / 254

Pool other :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.6.1           10.128.6.1       - 10.128.6.254      2    / 8     / 254
nsk-donskaya-vabazlov-gw-1#
```

Получение адресов

```
nsk-donskaya-vabazlov-gw-1#
nsk-donskaya-vabazlov-gw-1#sh ip dhcp bin
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.128.3.30      0090.21E7.A67C           --                     Automatic
10.128.3.31      0001.9674.D7CE           --                     Automatic
10.128.4.30      00E0.B09A.6776           --                     Automatic
10.128.5.30      0001.C97D.3A0B           --                     Automatic
10.128.6.30      0001.6472.D003           --                     Automatic
10.128.6.31      0001.42BA.6960           --                     Automatic
nsk-donskaya-vabazlov-gw-1#
```

Рис. 7: DHCP client

Проверка связи



Работа DHCP (Simulation)

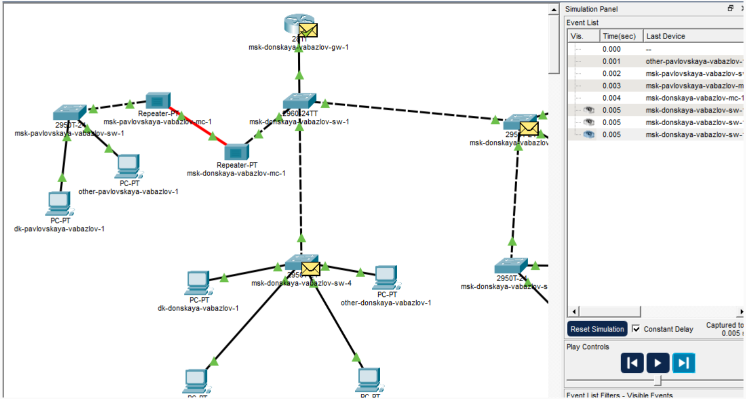
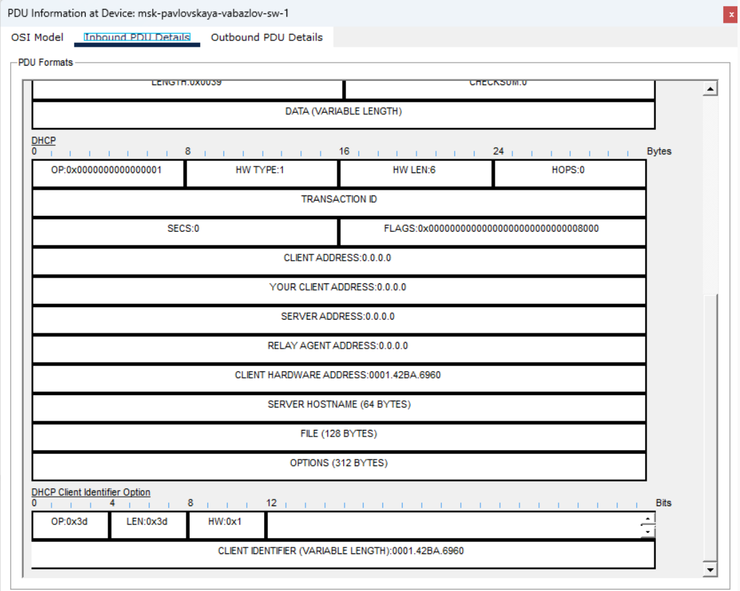


Рис. 9: Simulation

DHCP Discover



DHCP Offer

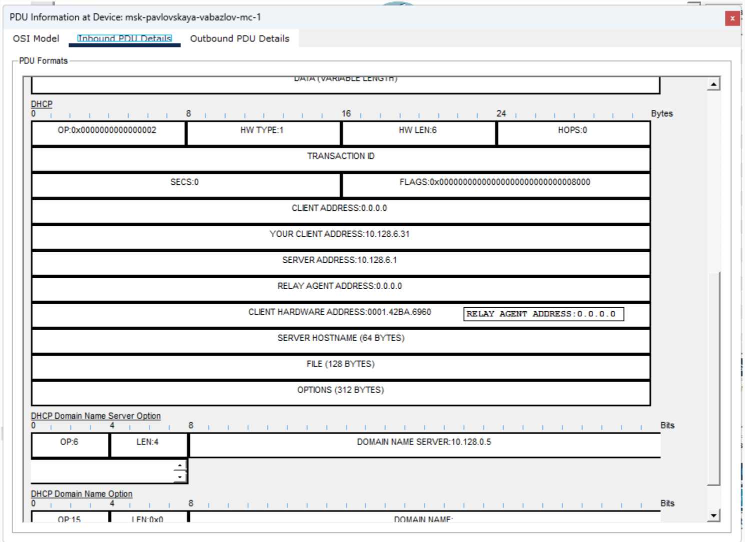


Рис. 11: Offer

DHCP Request

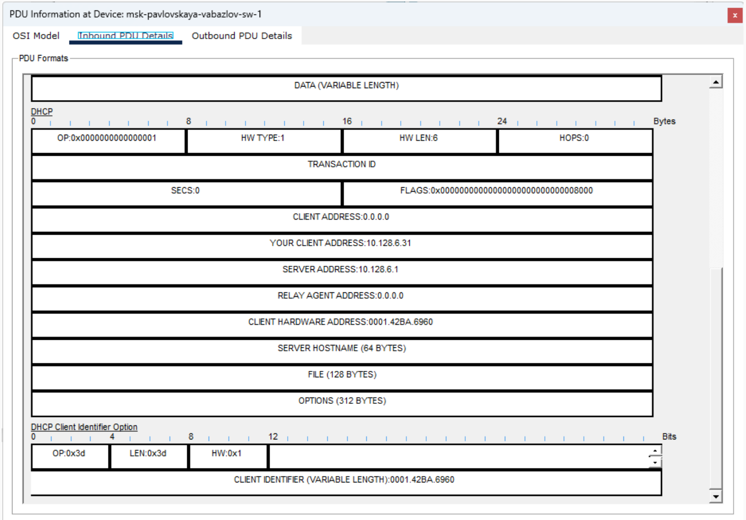
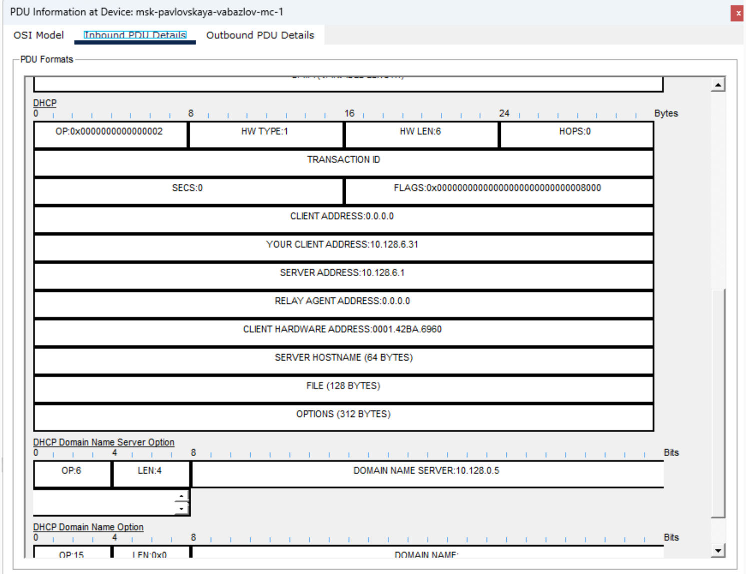


Рис. 12: Request

DHCP ACK


PDU Information at Device: msk-pavlovskaya-vabazlov-mc-1
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
DHCP
0
8
16
24
Bytes
OP:0x0000000000000002
HW TYPE:1
HW LEN:6
HOPS:0
TRANSACTION ID
SECS:0
FLAGS:0x0000000000000000000000000008000
CLIENT ADDRESS:0.0.0.0
YOUR CLIENT ADDRESS:10.128.6.31
SERVER ADDRESS:10.128.6.1
RELAY AGENT ADDRESS:0.0.0.0
CLIENT HARDWARE ADDRESS:0001.42BA.6960
SERVER HOSTNAME (64 BYTES)
FILE (128 BYTES)
OPTIONS (312 BYTES)
DHCP Domain Name Server Option
0
4
8
Bits
OP:6
LEN:4
DOMAIN NAME SERVER:10.128.0.5
DHCP Domain Name Option
0
4
8
Bits

Итоги работы

Вывод

- настроен DNS-сервер
- реализовано разрешение имён
- настроен DHCP для нескольких подсетей
- проверена автоматическая выдача адресов
- исследован процесс DHCP в Simulation

Получены навыки настройки сетевых сервисов
и анализа работы протоколов DHCP и DNS